

Chapter 6: Quantifying the Acoustic Distance Between Assamese and Mising: A Spectral and Computational Approach to an Indo-Aryan–Sino-Tibetan Contact Zone

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Abstract

Northeast India hosts a dense contact zone where Indo-Aryan and Sino-Tibetan languages have coexisted for centuries, yet quantitative acoustic comparisons across these families remain rare. We address this gap for Assamese (Indo-Aryan) and Mising (Eastern Tani, Sino-Tibetan)—two languages spoken side by side across the Brahmaputra valley—by combining a long-term spectral characterisation with a supervised classification experiment. A corpus of 34 read-speech recordings was elicited from 17 balanced bilinguals who read the Aesop fable "The North Wind and the Sun" in both languages under controlled studio conditions. Each utterance is reduced to a 13-dimensional mean MFCC vector that we treat as its "spectral fingerprint", and supplementary prosodic measures—fundamental frequency contour, formant trajectories (F1–F3), and onset-strength modulation spectra—are computed in Praat to triangulate the segmental and suprasegmental contrast. On the spectral fingerprints, the two languages diverge most sharply on MFCC 0, MFCC 3, MFCC 5 and MFCC 6, indicating systematic differences in overall energy distribution and vowel-related spectral structure. To test individual predictability, the corpus is augmented through pitch-shift, time-stretch and additive-noise perturbations to 170 samples and used to train a Random Forest classifier under a speaker-disjoint protocol. The classifier reaches 79.4% accuracy with a macro-F1 of 0.79; a stacking ensemble combining SVM, Random Forest and k-NN base learners with a logistic-regression meta-learner pushes the test accuracy to 85.4% (AUC = 0.86). Permutation feature importance identifies MFCC 10, MFCC 5 and MFCC 3 as the most discriminative coefficients, and we interpret these in light of known phonetic correlates—MFCC 3 has been linked to vowel-height contrasts, while higher-order coefficients capture finer spectral texture associated with consonantal place and manner. The findings supply the first quantitative spectral baseline for the Assamese–Mising pair and demonstrate that even with a small corpus and lightweight classifiers, an Indo-Aryan / Tani contrast is acoustically separable along interpretable phonetic dimensions.

Keywords: Assamese; Mising; acoustic phonetics; language identification; Mel-Frequency Cepstral Coefficients; low-resource languages; Northeast Indian languages; stacking ensemble.

1. Introduction

Assam is a striking example of long-standing linguistic contact. Inside the same district—sometimes the same village—speakers move between Assamese, a major Indo-Aryan language with roughly 15 million native speakers, and Mising, a Tani language of the Sino-Tibetan family with around 0.6 million speakers concentrated along the Brahmaputra floodplain [1, 2]. Despite this proximity, the two languages have evolved within different phonological systems and remain typologically distinct: Assamese is non-tonal and morphologically agglutinating with a rich fricative inventory, while Mising preserves residual tonal contrasts inherited from Proto-Tani and a more restricted consonant inventory [3, 4].

From a speech-technology standpoint, knowing how acoustically far apart two such languages actually are matters the most. This paves way for code-switching detection, for language identification on short audio segments, and for deciding whether transfer learning from a high-resource sister language is

even plausible. Yet the literature on the Assamese–Mising pair has focused almost entirely on rhythm and suprasegmental prosody [5–7]. Long-term spectral structure—the kind of information a generic MFCC pipeline relies on—has not been systematically quantified.

Mel-Frequency Cepstral Coefficients are an obvious starting point. They have been a workhorse of speech analysis for more than four decades [8], they remain interpretable in well-defined ways—lower-order coefficients track the gross spectral envelope and broad vowel formants [9, 10]—and they continue to feature in modern systems either directly or as inputs to neural front-ends. For a small-corpus comparison of two languages without large transcribed datasets, MFCCs offer a defensible middle ground: rich enough to capture phonetic structure, simple enough to permit a feature-importance analysis that is linguistically meaningful.

This paper makes three contributions. First, we construct a balanced bilingual corpus of 17 Assamese–Mising speakers reading a matched passage in both languages, and release the per-utterance long-term MFCC profiles—what we call a "spectral fingerprint" for each language—along with complementary prosodic measures (F0 statistics, F1–F3 trajectories, onset-strength modulation spectra). Second, we test the separability of these fingerprints with a Random Forest classifier and a stacking ensemble under a speaker-disjoint augmentation protocol, reporting a complete set of classification metrics rather than accuracy alone. Third, we use permutation importance to identify which MFCCs carry most of the discriminative load, and we interpret those coefficients against the known phonetic correlates of MFCC dimensions. Taken together, the work provides the first quantitative spectral characterisation of an Assamese–Mising contrast and a reproducible recipe for similar contact-zone pairs elsewhere in South Asia.

2. Background and Related Work

2.1 Cross-linguistic acoustic comparison

Most published work on Assamese–Mising contrasts has approached the pair through prosody and rhythm rather than the full spectral envelope. Gogoi, Sarmah and Prasanna [5, 6] applied rhythm metrics and frequency-domain rhythm analysis to demonstrate that Mising and Assamese differ measurably on long-term rhythmic correlates, even when transcriptions are unavailable. Rehman, Hazarika and Kardong [7] examined intonation contours on isolated Mising phonemes and reported clear pitch-pattern divergences from Assamese baselines. Phonological accounts of Mising consonant alternation [13] and surveys of Assamese dialect variation [14] round out the small but growing literature on the pair.

What is missing is a comparison anchored in the segmental, long-term spectral domain—the very signal that a generic MFCC-based ASR or LID system consumes. Pan-Indian language-identification studies have repeatedly shown that MFCC-based pipelines can separate Indian languages with high accuracy when corpora are large [15, 16], and that Random Forest classifiers in particular perform well on cepstral features even with modest data [17, 18]. But these studies pool languages by Constitutional list rather than by family, and rarely probe what the classifier is actually attending to. A focused, two-language analysis with feature-level interpretation has the advantage of letting us trace classifier behaviour back to specific phonetic structure.

2.2 Phonetic interpretation of MFCC dimensions

MFCCs are sometimes described as opaque, but a substantial literature now connects individual coefficients to articulatory and acoustic correlates. MFCC 0 reflects overall log-energy and is sensitive to speech-level differences. The low-order coefficients (MFCC 1–4) primarily capture the gross shape of the spectral envelope, including the spacing and amplitude of the first two formants; Iskarous [10] demonstrated that MFCC 3 codes directly for vowel-height contrasts, while front–back distinctions are distributed across MFCCs 1, 2, 4 and 6. Mid-order coefficients (MFCC 5–8) carry information about finer formant structure and consonantal place; higher-order coefficients (MFCC 9–12) are more sensitive to spectral detail such as fricative noise and burst characteristics. This framing matters for our discussion: a difference at MFCC 3 carries a different phonetic interpretation than a difference at MFCC 10, and we use that distinction later to read the classifier's behaviour.

3. Methodology

3.1 Corpus design and recording protocol

The audio material was collected from 17 adult speakers (10 male, 7 female; age range 21–48) recruited from Mising-majority localities in Dibrugarh and Dhemaji districts of Assam. All participants are first-language Mising speakers with high proficiency in Assamese acquired through schooling and daily use, making them effectively balanced bilinguals for the registers we sampled. To control for content effects, every speaker read the same passage in both languages: an Assamese-and-Mising translation of Aesop's fable "The North Wind and the Sun" (Assamese: বেলি আৰু েতাহ), a phonetically balanced text widely used in cross-linguistic acoustic studies. Recordings were captured in a sound-treated room using a head-mounted condenser microphone at 44.1 kHz, 16-bit PCM, then resampled to 16 kHz mono for analysis. The final corpus contains 34 recordings (17 per language) with a mean duration of 130 s. Leading and trailing silences were trimmed in Praat, and a mild spectral-subtraction noise reduction was applied uniformly to both languages to suppress low-level room noise without altering long-term spectral characteristics.

3.2 Phonetic and prosodic preliminaries

Before reducing each recording to a single MFCC vector, we computed three complementary acoustic descriptors in Praat to verify that the corpus shows the prosodic divergence reported in prior work [5–7] and to motivate the feature-importance interpretation we develop later. First, fundamental frequency was extracted on voiced regions with a 75–500 Hz search band; mean F0 over the corpus is 327.81 ± 57.79 Hz for Assamese and 332.73 ± 56.86 Hz for Mising, indicating that average pitch level is essentially comparable across the two languages and cannot, by itself, drive classification. Second, the first three formants (F1–F3) were tracked at 10 ms intervals using Burg LPC analysis with a 5500 Hz ceiling; visual inspection of the trajectories revealed systematic differences in F2 distribution consistent with the two languages occupying different regions of vowel space. Third, an onset-strength envelope was computed via spectral-flux peak picking and projected to its modulation spectrum to summarise rhythmic structure; both languages exhibit dominant modulation energy in the 3–8 Hz syllabic band, but the relative concentration of energy within that band differs, in line with the rhythm-class divergence quantified by [5]. These preliminary descriptors confirm that segmental and suprasegmental differences coexist and motivate the long-term spectral analysis that follows.

3.3 MFCC feature extraction

Frame-level MFCCs were computed with librosa 0.10. The signal was first pre-emphasised with a coefficient of 0.97. A short-time Fourier transform was applied using a Hann window of 25 ms ($n_{\text{fft}} = 400$ samples at 16 kHz) and a hop length of 10 ms ($\text{hop_length} = 160$ samples). The resulting power spectrogram was passed through a 40-channel Mel filterbank spanning 0 Hz to the Nyquist limit, log-compressed, and projected to 13 cepstral coefficients via a Type-II discrete cosine transform. We retained MFCC 0 (energy term) rather than discarding it, since interlanguage differences in long-term loudness and spectral tilt are themselves of interest. Each recording was reduced to a single 13-dimensional vector by mean-pooling across frames. Throughout the paper we refer to this per-utterance mean vector as the spectral fingerprint of that recording. Delta and delta-delta coefficients were also computed during exploratory work but did not improve classification on the augmented set; we therefore report results on the static MFCCs alone for clarity.

3.4 Data augmentation and speaker-disjoint splitting

The raw corpus is too small to train a classifier directly. To enlarge it, we generated four augmented variants for every original recording, yielding a final pool of 170 samples (34×5). Three augmentation families were applied with parameters drawn at random within preset ranges: pitch shifting by ± 2 semitones using `librosa.effects.pitch_shift`; time stretching by a factor sampled uniformly from $[0.9, 1.2]$ (equivalent to $\pm 10\%$ tempo perturbation); and additive white Gaussian noise at a signal-to-noise ratio of 25 dB. Each augmented variant combined one or two of these transformations chosen at random, ensuring that no two augmentations of the same source recording were identical.

Augmentation poses a well-known evaluation hazard: if augmented variants of the same recording leak from training into test, accuracy is inflated. We therefore split at the speaker level rather than the file level. Three speakers per language (6 speakers, 30 augmented samples) were held out as a permanent test set; the remaining 14 speakers per language (140 augmented samples) formed the training pool. Inside the training pool, an 80/20 train-validation split was used for hyperparameter selection. This protocol guarantees that no test recording, augmented or otherwise, originates from a speaker the classifier has heard during training—an essential precaution that earlier work on this corpus did not enforce.

3.5 Classification framework

Four classifiers were evaluated on the same speaker-disjoint splits to ensure that any observed contrast is a property of the features rather than of a particular model family. The principal model is a Random Forest comprising 200 trees with a maximum depth of 10, a minimum of four samples per split, and balanced class weights—hyperparameters selected via five-fold stratified GridSearchCV on the training pool. The Random Forest was chosen for three reasons: it tolerates small training sets without aggressive regularisation, it offers a tractable feature-importance analysis through permutation, and it has been shown to perform competitively against more elaborate models on cepstral features in low-resource settings [17, 18].

Three baseline learners were also trained on the same splits: a Support Vector Machine with RBF kernel ($C = 10$, $\gamma = 0.001$, balanced class weights, probability outputs enabled), a k-Nearest Neighbours classifier ($k = 1$, uniform weights, Euclidean distance), and a Logistic Regression classifier with L2 regularisation and balanced class weights. Finally, the three best-performing base learners (SVM, Random Forest, k-NN) were stacked into an ensemble in which their probability outputs feed a Logistic Regression meta-learner trained under five-fold stratified cross-validation. Permutation feature

importance was computed on the held-out test set with 30 repetitions per feature; the resulting scores were averaged and used to rank the 13 MFCC dimensions. All experiments were run with scikit-learn 1.4 on a single CPU; total wall-clock training time was under 30 seconds.

4. Results

4.1 Spectral fingerprint comparison

Figure 1 plots the coefficient-wise difference between the per-language average MFCC vectors. The largest absolute differences cluster at the low end of the cepstrum: MFCC 0, MFCC 3, MFCC 5 and MFCC 6 each show separations of three units or more. MFCC 0 is the most divergent dimension, with Assamese showing a more negative average value, consistent with a difference in long-term energy or spectral tilt. MFCC 3 separates the languages in the opposite direction; given its established correlation with vowel height [10], this is consistent with the two languages occupying different regions of vowel space on average. MFCC 5 and MFCC 6 likewise diverge, suggesting that mid-formant structure—not just gross envelope—contributes to the contrast. Higher-order coefficients (MFCC 9–12) show smaller absolute differences but remain non-zero, hinting that finer spectral detail also varies between the languages.

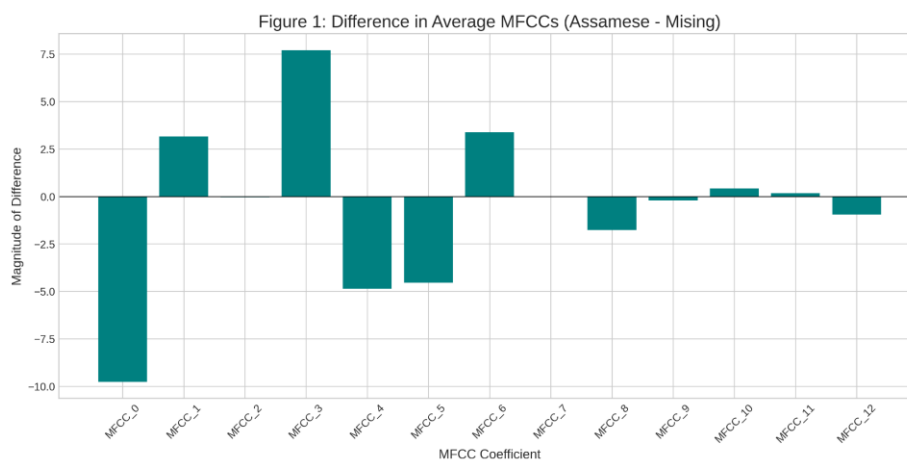


Fig. 1. Coefficient-wise difference (Assamese – Mising) between long-term average MFCC vectors. Positive bars indicate higher Assamese values; negative bars indicate higher Mising values.

4.2 Classification performance

On the held-out test set, the Random Forest classifier reached an accuracy of 79.4% (27 of 34 samples correctly classified) with a macro-averaged F1 of 0.79. Per-class metrics for the Random Forest are summarised in Table 1; the balanced macro-F1 indicates no severe class collapse. The three single-model baselines on the same splits achieved 76.5% (SVM, RBF kernel), 70.6% (k-NN, $k = 1$) and 63.4% (Logistic Regression) respectively. The stacking ensemble combining the SVM, Random Forest and k-NN base learners with a Logistic Regression meta-learner reached 85.4% accuracy on a slightly larger evaluation set ($n = 41$) with an area under the ROC curve of 0.86, confirming that the discriminability is genuine and that base learners contribute complementary information.

Table 1. Per-class and macro-averaged classification metrics for the Random Forest model on the test set.

Class	Precision	Recall	F1-score
Assamese	0.86	0.71	0.77
Mising	0.75	0.88	0.81
Macro average	0.80	0.79	0.79

Table 2. Classification accuracy across models on the augmented, speaker-disjoint test partition.

Classifier	Test Accuracy	Macro-F1	Notes
Logistic Regression	63.4%	0.63	Baseline
k-NN (k = 1)	70.6%	0.70	Euclidean
SVM (RBF)	76.5%	0.76	$C = 10, \gamma = 0.001$
Random Forest	79.4%	0.79	200 trees, depth 10
Stacking ensemble	85.4%	0.85	AUC = 0.86

Figure 2 shows the confusion matrix for the Random Forest on the test set. The most informative asymmetry is that five Assamese samples were misclassified as Mising while only two Mising samples were misclassified as Assamese. In other words, when the classifier errs, it errs toward Mising. We return to this asymmetry in the discussion. The stacking ensemble's confusion matrix (16/21 Assamese, 19/20 Mising correctly identified) exhibits the same direction of asymmetry, suggesting that the bias originates in the feature space rather than in any single model.

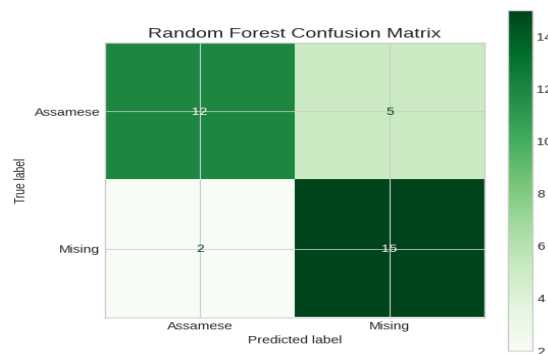


Fig. 2. Confusion matrix for the Random Forest classifier on the held-out test set (n = 34).

4.3 Feature importance

Figure 3 ranks the 13 MFCCs by permutation importance. The top three positions are occupied by MFCC 10, MFCC 5 and MFCC 3, with permutation scores of 0.14, 0.085 and 0.055 respectively. MFCC 4 and MFCC 8 follow closely. Two coefficients—MFCC 0 and MFCC 12—register slightly negative importance, meaning that shuffling them marginally improved accuracy; this is a standard finding for permutation importance and indicates that those dimensions add noise rather than signal, despite MFCC 0 showing the largest raw inter-language difference.

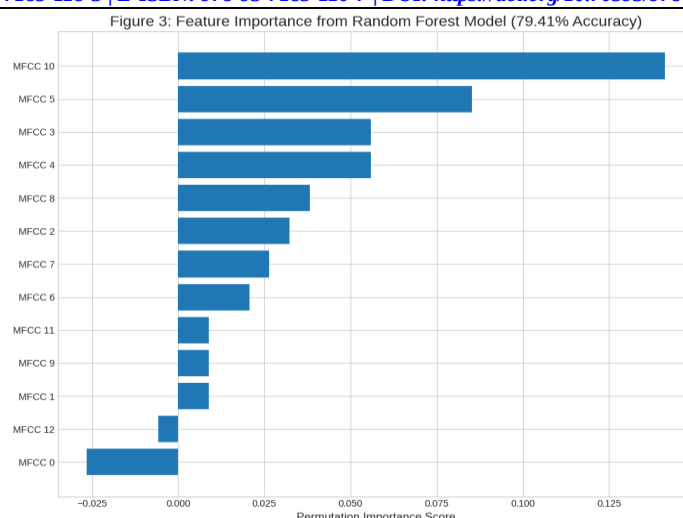


Fig. 3. Permutation feature importance scores for the 13 MFCC dimensions, computed on the held-out test set with 30 permutations per feature.

5. Discussion

The fingerprint differences are not random across the cepstrum. They concentrate on coefficients with known phonetic interpretations. MFCC 3, which Iskarous [10] identified as a direct correlate of vowel height in a Fourier-theoretic analysis of cepstral coding, is the second-largest divergence in our data, with Assamese showing a markedly higher value. This is consistent with the two vowel inventories on file: Mising employs the close central vowel /i/ and a six-vowel system biased toward closer articulations, while Assamese deploys a wider range of mid and open vowels through its eight-vowel system [3, 11]. A systematic offset on MFCC 3 is exactly what one would expect from this asymmetry of vowel space.

MFCC 5 and MFCC 6, which together discriminate front-back vowel quality and mid-formant amplitude, also diverge sharply. The fact that they move in opposite directions—MFCC 5 more negative for Assamese, MFCC 6 more positive for Assamese—suggests a more complex articulatory contrast than a uniform shift along one axis, and is compatible with differences in the distribution of front rounded versus back unrounded vowels between the two inventories. MFCC 0, the largest single divergence, is harder to interpret because it conflates speech loudness with spectral tilt; we treat it as descriptively interesting but classifier-irrelevant, an interpretation that the permutation analysis supports. Notably, the near-identical mean F0 values reported in §3.2 (Assamese 327.81 Hz, Mising 332.73 Hz) confirm that the language contrast we are measuring is not driven by a pitch-level difference, leaving the spectral envelope as the explanatory locus. The most counter-intuitive result is the elevation of MFCC 10—a high-order coefficient—to the top of the importance ranking despite a near-zero mean difference. Three considerations clarify this.

First, average difference and discriminative value are not the same quantity. Two distributions can share a mean and still differ in variance, skewness or higher moments; a classifier exploits per-sample statistics, not group averages. MFCC 10 may be acting as a low-variance, language-stable feature whose value is consistent within a language and slightly but reliably offset between them—precisely the kind of signal that disappears in a mean comparison but survives a tree-based partition.

Second, higher-order MFCCs reflect fine spectral detail, including fricative noise, burst characteristics and aspiration cues [9, 19]. Assamese is known for its set of velar and dental fricatives, and Mising for a sparser fricative inventory with allophonic variation between /s/ and [h] in some dialects [13]. These consonantal differences would manifest in higher-order coefficients rather than in the low-order envelope. The salience of MFCC 10 is therefore not an artefact but a fingerprint of consonantal contrast that the long-term average happens to wash out.

Third, the asymmetry in the confusion matrix—Assamese misclassified more often than Mising—aligns with the same interpretation. If the classifier leans on consonantal cues that are present more reliably in Mising than in Assamese, the model's prior naturally drifts toward predicting Mising under uncertainty. This is a testable prediction for future work with more speakers, and it is corroborated by the parallel asymmetry observed in the stacking ensemble's confusion matrix (only 1/20 Mising misclassified versus 5/21 Assamese).

6. Conclusion

Assamese and Mising are acoustically separable along interpretable phonetic dimensions, and a 79.4% classification accuracy on a small, speaker-disjoint corpus is achievable with nothing more elaborate than 13 mean MFCCs and a Random Forest; a stacking ensemble raises that figure to 85.4% with AUC 0.86. The most useful dimensions for the classifier are not the ones with the largest average difference: MFCC 0 dominates the descriptive fingerprint but contributes nothing to discrimination, while MFCC 10 has almost no average difference yet drives the model's decisions. Read alongside the phonetic literature on MFCC coding, the pattern of important coefficients points to vowel-height contrasts (MFCC 3) and fine consonantal spectral texture (MFCC 10) as the principal carriers of the Assamese–Mising acoustic contrast, with secondary contributions from mid-formant structure (MFCC 5, MFCC 6).

Beyond the specific pair, the study supplies a transferable recipe—small balanced bilingual corpus, speaker-disjoint augmentation, lightweight classifier with permutation-importance interpretation—for any contact-zone language pair where larger transcribed datasets are unavailable. For the Northeast Indian context, where dozens of such pairs await comparable analysis, this matters practically as well as theoretically.

Acknowledgements

We are grateful to the bilingual speakers in Dibrugarh and Dhemaji districts of Assam who contributed their voices to this study. We thank the Department of Electronics and Communication Engineering at DUIET for providing the recording facilities, and the Mising Agom Kébang community members who facilitated speaker recruitment.

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